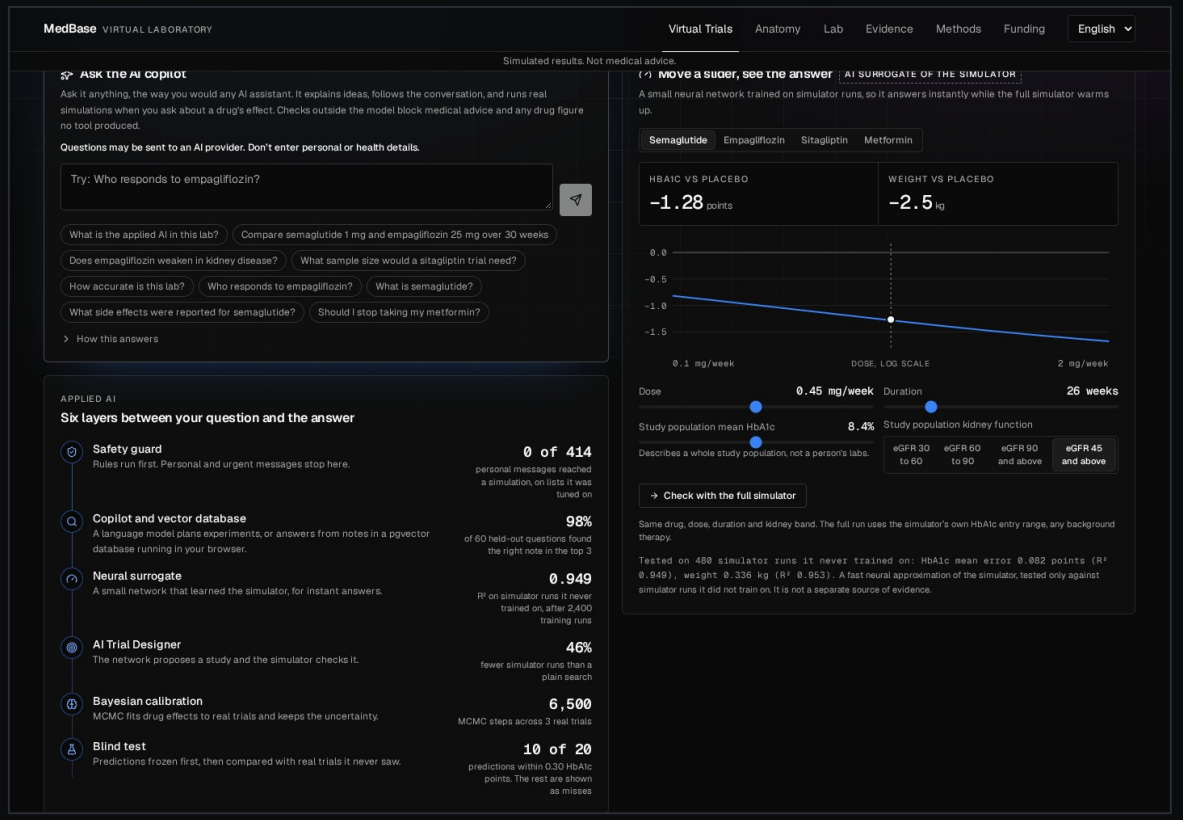
INFINEX 2K26 | APPLIED AI

MedBase Virtual Trials Lab

Run a simulated type 2 diabetes drug trial in your browser, on virtual patients built from real health survey data, and see how far to trust the answer.



Try it live
https://zade06079-medbase-virtual-trials-lab.static.hf.space

Simulated results. Not medical advice.

The problem: trials can't be rehearsed, and health AI can't be checked

The rehearsal gap

Drug trials cost years and enormous sums. Students and early researchers have no accessible way to explore how a study would behave before it exists, and most open diabetes simulators cover type 1 only.

The trust gap

Many health AI tools give confident answers with no way to check them. A learner can't tell a sourced figure from a plausible one, or see where the tool has been wrong before.

Our answer: virtual patients from real survey data, AI that plans and explains, and a simulator that computes every efficacy number.

The solution: ask in plain English, and a real simulator answers

1

Your question

Plain English, such as "How much does semaglutide lower HbA1c?"

2

Safety guard

Personal and urgent messages stop here, before any AI

3

AI copilot

Plans the experiment, or finds notes by meaning

4

Simulator

Computes every efficacy number on your own device

5

Checked answer

Code checks the figures, and the track record sits beside them

Models propose and explain. The simulator computes.

Key features

Conversational AI copilot

Explains ideas, follows up on "why?", and runs real simulations when you ask about a drug's effect.

Virtual trials

Four real drugs and hypothetical molecules, on virtual patients built from real survey labs.

AI Trial Designer

Set a goal. A neural network proposes a study, and the simulator checks it step by step.

Instant neural sliders

Move dose or duration and get an answer in milliseconds, then race it against the full simulator.

Vector database in the browser

PostgreSQL with pgvector finds the right notes by meaning, with no server and no account.

Public blind test

Every frozen prediction beside the real result, including the ones it got wrong.

Technology stack

AI models

- NVIDIA Nemotron 3.5 on NIM: language model
- nemotron-3-embed-1b: embeddings
- Neural surrogate: a small MLP
- Bayesian MCMC with emcee

Science

- Python, NumPy, SciPy
- scikit-learn, statsmodels
- NHANES survey data
- Published trial results

In the browser

- React 19, TypeScript, Vite, Tailwind
- Pyodide: the Python simulator
- PostgreSQL and pgvector 0.8.7, through PGlite
- three.js: 3D body views

Hosting and safety

- Hugging Face Spaces: static site
- Vercel: proxy that holds the AI key
- Rule-based guard before any model
- Content Security Policy, key scans

Innovation: applied AI you can check

What is new

- A real simulator and a real vector database run inside the browser, at no server cost
- The language model plans and explains, while code checks its figures against what actually ran
- Predictions were frozen before real results were opened: 10 of 20 landed within 0.30 HbA1c points, and every miss stays public
- We attacked our own safety guard and published what got through

R^2 0.949

neural surrogate, on 480 simulator runs it never trained on

98%

of 60 held-out questions found the right note in the top 3

4 ms

median vector database query, under Node

Real-world impact

Who it helps

- Students and early researchers can rehearse a study design for free, with no account, on an ordinary laptop
- Learners see how far to trust an AI answer, because its track record sits beside it
- It costs almost nothing to run, because each visitor's browser does the simulating
- Next: pilot it with a class, and add Indian patient data

46%

fewer simulator runs than a plain search, across 36 study goals

0

servers to run for the simulator, and no account needed

What it is not, and where to try it

What it is not

- Not clinically validated, and not advice for any person
- In its blind test it missed its accuracy mark and did not clearly beat a simple rival
- No cardiovascular outcomes, hypoglycaemia or serious adverse events simulated
- Virtual patients come from a US survey, not an Indian cohort

Live demo https://zade06079-medbase-virtual-trials-lab.static.hf.space

Code and validation report https://github.com/shreynachaegari-glitch/medbase-virtual-trials-lab

Built on MedBase by Team Aegis.